Индивидуальное задание

16) 1. $\lim\limits_{n\to\infty} \frac{7n-3}{8-7n} = -1$

$\forall \varepsilon > 0 \quad \exists n_0 \in \mathbb{N} : \forall n > n_0 \quad \left| \frac{7n-3}{8-7n} + 1 \right| < \varepsilon$

$\left| \frac{7n-3}{8-7n} + 1 \right| = \left| \frac{7n-3+8-7n}{8-7n} \right| = \frac{5}{7n-8} < \varepsilon$

$7n - 8 > \frac{5}{\varepsilon}$

$n > \frac{5+8\varepsilon}{7\varepsilon}$

$n_0 = \left[\frac{5+8\varepsilon}{7\varepsilon} \right] = \left[\frac{5}{7\varepsilon} + \frac{8}{7} \right]$

2. $\lim\limits_{n\to+\infty} \left(3 - \sqrt{n \ln n}\right) = -\infty$

$\forall \varepsilon > 0 \quad \exists n_0 \in \mathbb{N} : \forall n > n_0 \quad \left(3 - \sqrt{n \ln n}\right) < -\frac{1}{\varepsilon}$

$n \ln n > \left(3 + \frac{1}{\varepsilon}\right)^2$

$n \ln n > n > \left(3 + \frac{1}{\varepsilon}\right)^2$

$n_0 = \left[\left(3 + \frac{1}{\varepsilon}\right)^2 \right]$

$\lim\limits_{n\to\infty} \frac{\sqrt{n+1}\,\operatorname{arctg} n}{2n+5} = 0$ $\quad \forall \varepsilon > 0 \ \exists n_0 \in \mathbb{N} : \forall n > n_0$

$\left| \frac{\sqrt{n+1}\,\operatorname{arctg} n}{2n+5} \right| < \varepsilon$

$\left| \frac{\sqrt{n+1}\,\operatorname{arctg} n}{2n+5} \right| = \frac{\sqrt{n+1}\,\operatorname{arctg} n}{2n+5} < \frac{\sqrt{n+1}\cdot\frac{\pi}{2}}{n} < \varepsilon$

уменьшили $\operatorname{arctg} n \Rightarrow$ ув. дроб

$\frac{n}{\sqrt{n+1}\cdot\frac{\pi}{2}} > \frac{1}{\varepsilon}$

$\frac{1}{\frac{\pi}{2}} \sqrt{\frac{n^2}{n+1}} > \frac{1}{\frac{\pi}{2}} \sqrt{\frac{n^2-1}{n+1}} = \frac{1}{\frac{\pi}{2}} \sqrt{\frac{(n-1)(n+1)}{(n+1)}} = \frac{\sqrt{n-1}}{\frac{\pi}{2}} > \frac{1}{\varepsilon}$

$n > \frac{\pi^2}{4\varepsilon^2} + 1$

$n_0 = \left[\frac{\pi^2}{4\varepsilon^2} + 1\right]$

$\boxed{n_0 = \left[\frac{\pi^2}{4\varepsilon^2} + 1\right]}$